

Business Economic and Financial Data

Analysis of Player Numbers Fluctuation Using Time Series Models

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Background

Player acquisition is only one stage of commercial success in the video game industry. Developers must also maintain engagement after launch, reactivate inactive players, and extend a product's commercial lifecycle. This report compares three successful games that followed different strategies: an Early Access and community-building approach for Baldur's Gate 3 (BG3), a major expansion strategy for Elden Ring (EDR), and long-term content support for Sid Meier's Civilization VI (CV6). The analysis evaluates how these strategies are reflected in player activity and what other developers can learn from them.

Data Description

The analysis uses SteamDB peak-player data for three games. These observations include both new and returning users and therefore represent player activity rather than verified daily sales. To make the three datasets comparable with publicly reported sales totals, the original peak-player series was rescaled using a fixed game-specific factor. The transformed values should be interpreted as an estimated activity measure, not as directly observed daily new-player counts.

Methodology

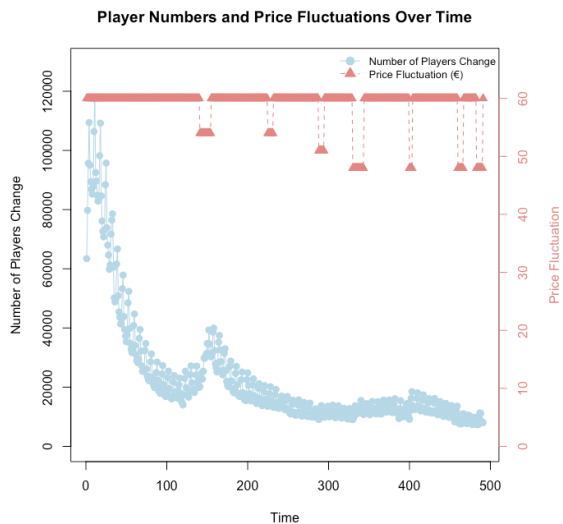


Figure 1.1

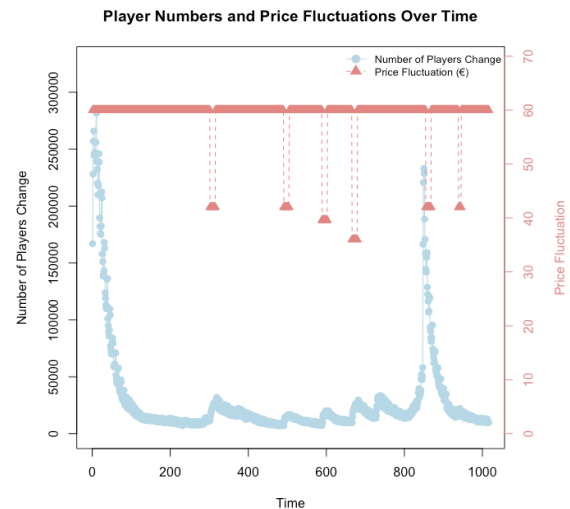


Figure 1.2

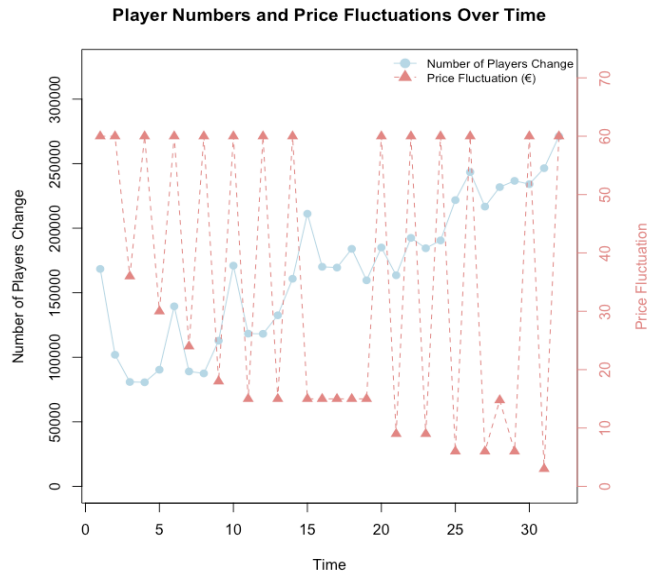


Figure 1.3

Figure 1.1 The player number fluctuation of Baldur's Gate 3 daily. Figure 1.2 The player number fluctuation of Elden Ring daily. Figure 1.3 The player number fluctuation of Sid Meier's Civilization quarterly.

The three series have nonlinear lifecycle patterns, including rapid post-launch decline, event-driven peaks, and long-term growth. Diffusion models were therefore used to describe adoption dynamics, while Prophet and ARIMA-based models were used for forecasting and short-term variation. Baldur's Gate 3 and Elden Ring were analysed with a Generalized Bass Model containing event shocks and with Prophet forecasts. Civilization VI was aggregated to quarterly frequency and analysed using the standard Bass Model, ARIMA, and a Bass-plus-SARMAX refinement.

Figure 1.1 is the plot of the player number for Baldur's Gate 3, characterized by an extreme initial peak followed by a smaller peak. The general Bass model is not appropriate for this dataset, employing a Generating Bass model with two exponential shocks provides insightful results. Using trend method or exponential smoothing method are not ideal due to the rapid decrease in numbers post peak, whereas the Prophet performs well, particularly incorporating shocks into the model.

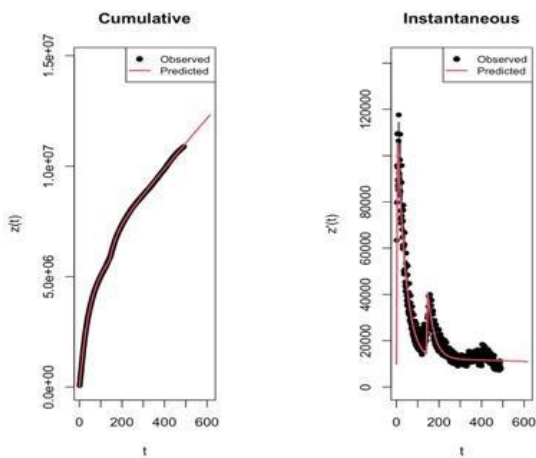
Figure 1.2 shows the player data for Elden Ring. It uses the same GBM for analysis and the Prophet model for forecasting as used in BG3. However, the plot has two

shocks spaced over a long interval. In the GBM analysis process, splitting the dataset into two parts gets a more accurate analysis.

Figure 1.3 features Sid Meier's Civilization VI, which differs from figure 1.1 and 1.2, because the game was released much earlier, about nine years ago. Analyzing daily data can involve many fluctuations; by converting the daily changes in player numbers to quarterly data using Python, the plot exhibits distinct characteristics comparing with other two. The general BM provides a good fit for this dataset, the ARIMA model is used for additional insights. Ultimately, a more sophisticated SARIMAX model is applied for forecasting.

Analysis

Baldur's Gate 3



Residuals:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	-49033.7	-21134.1	-4390.5	999.1	25881.3	102320.7

Coefficients:

	Estimate	Std. Error	Lower	Upper	p-value
m	2.538396e+07	2.778939e+06	1.993734e+07	3.083059e+07	1.82e-18 ***
p	3.883796e-04	3.855577e-05	3.128117e-04	4.639475e-04	8.71e-22 ***
q	9.317660e-04	1.777251e-04	5.834312e-04	1.280101e-03	2.37e-07 ***
a1	1.310047e+00	1.484412e-01	1.019108e+00	1.600987e+00	2.02e-17 ***
b1	-2.772393e-02	5.259540e-04	-2.875478e-02	-2.669308e-02	3.21e-202 ***
c1	1.075920e+01	5.812250e-01	9.620017e+00	1.189838e+01	3.47e-58 ***
a2	1.420010e+02	4.494600e-01	1.411201e+02	1.428819e+02	0.00e+00 ***
b2	-3.133196e-02	1.110184e-03	-3.350788e-02	-2.915603e-02	3.65e-104 ***
c2	2.681916e+00	6.372455e-02	2.557018e+00	2.806814e+00	1.59e-163 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29161.25 on 482 degrees of freedom
Multiple R-squared: 0.99988 Residual sum of squares: 409882477103

Figure 2.1

The full-period Generalized Bass Model captures the cumulative player-activity curve closely, with an in-sample R^2 of 0.99988. The innovation coefficient is $p = 0.000388$, while the imitation coefficient is $q = 0.000932$. Therefore, q is approximately 2.40 times p , indicating that the fitted diffusion pattern is more strongly associated with internal influence and player-to-player diffusion than with external adoption pressure. The instantaneous plot also captures the launch decline and the second major activity shock, although it does not reproduce every short-term fluctuation.

The launch-period activity reached approximately 117,000 in the transformed series, while the second major peak was close to 40,000. This means the later event generated roughly one-third of the initial peak. The result shows that post-launch content can reactivate a substantial portion of the audience, but it did not recreate the scale of the original release.

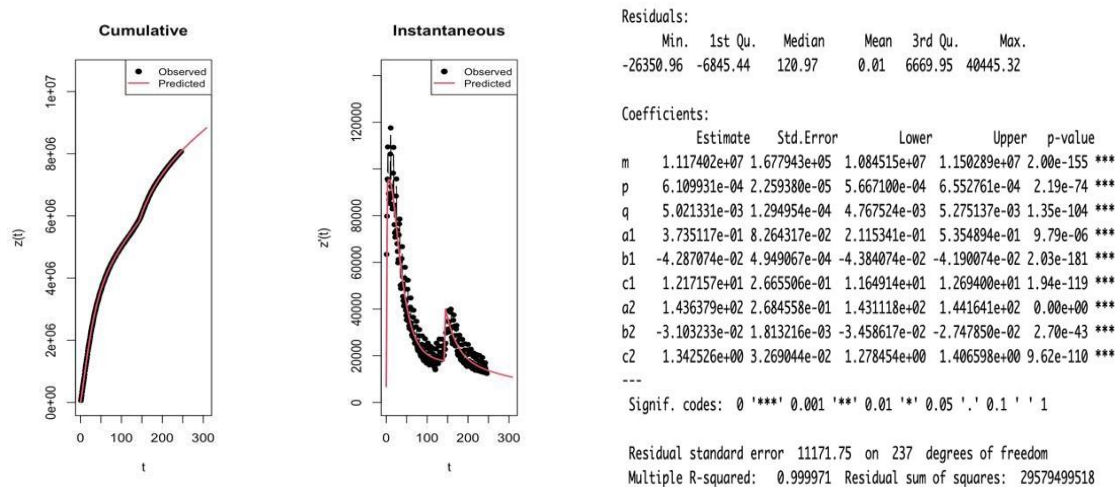


Figure 2.2

When the model is estimated on the first half of the dataset, the innovation coefficient is $p = 0.000611$ and the imitation coefficient is $q = 0.005021$. The q/p ratio increases to approximately 8.22, compared with 2.40 for the full period. The first-half model also produces a lower residual standard error of about 11,172, compared with 29,161 for the full-period model. These results suggest that community diffusion was especially strong during the launch and early post-launch stage.

For developers, this supports a community-first interpretation of the Early Access strategy. Early players can provide feedback, create online discussion, publish reviews, and generate visible social proof before and immediately after the full release. However, this approach is high risk: if the public version is technically unstable or poorly received, the same rapid community diffusion can spread negative impressions and weaken the full launch.

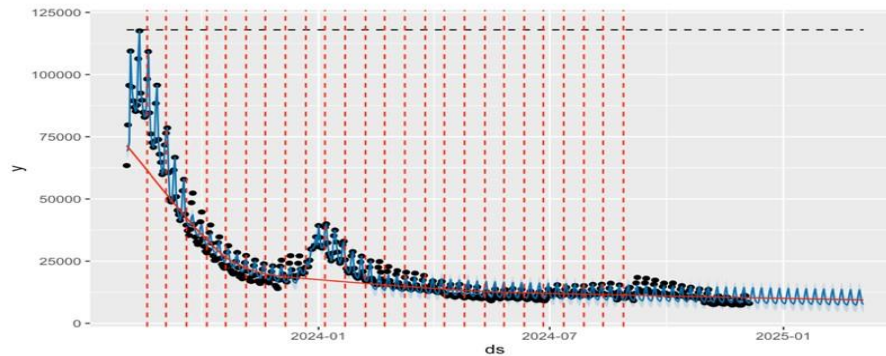


Figure 2.3

The selected Prophet model forecasts a stable but weekly fluctuating activity level over the following 90 days. During the final six forecast dates, point estimates range from approximately 7,447 to 13,416, with the highest values occurring on the weekend. The prediction intervals remain positive, and the overall trajectory is broadly flat after the major launch and January peaks. This indicates that, without another major content event, player activity is expected to remain at a lower but persistent baseline.

The business implication is that discounts alone are unlikely to recreate launch-scale engagement. The price plot shows several reductions, but the associated activity changes are much smaller than the launch and major content shock. A stronger retention plan would combine meaningful content updates, community communication, and promotional pricing rather than relying on discounts as the principal driver.

Elden Ring

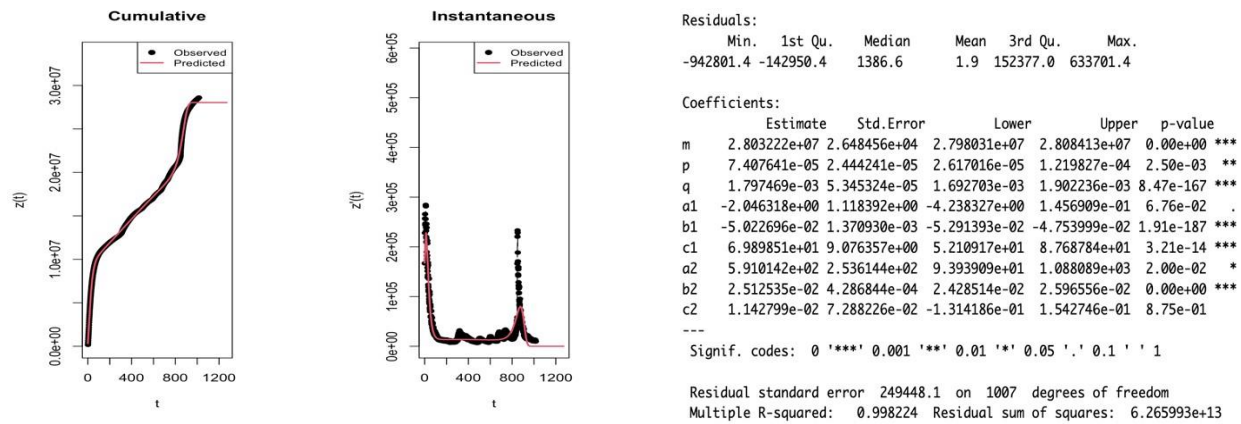
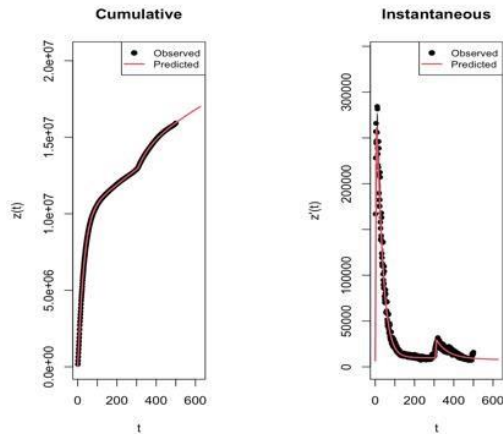


Figure 2.4

Elden Ring exhibits two dominant commercial events: the original launch and the release of Shadow of the Erdtree. The transformed activity measure reached about 284,000 around the initial launch and approximately 230,000 around the expansion, meaning the DLC peak was roughly 81% of the original launch peak. In the source player-count figures reported in the project, activity increased from 194,570 on 19 June 2024 to 558,139 on 20 June and reached 781,261 on 22 June. This is an increase of approximately 3 times from 19 June to the peak.

The full-period GBM has $p = 0.0000741$ and $q = 0.001797$, giving $q/p \approx 24.3$ and an in-sample R^2 of 0.998224. The cumulative pattern is captured well, but the instantaneous fit underestimates the sharp DLC peak and several smaller fluctuations. Therefore, the model supports a strong community-diffusion interpretation but is less suitable for reproducing the exact magnitude of exceptional event-driven peaks.



Residuals:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	-71652	-15840	-246	1661	20703	62182

Coefficients:

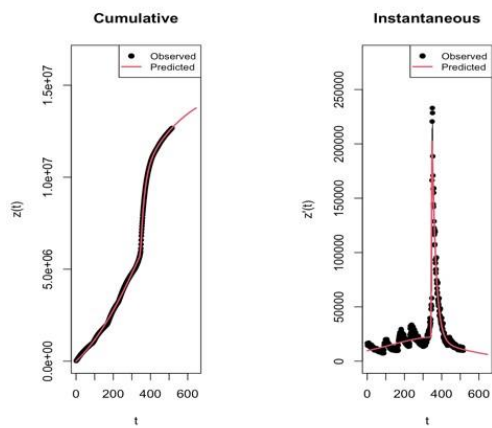
	Estimate	Std. Error	Lower	Upper	p-value
m	2.525670e+07	9.654883e+05	2.336438e+07	2.714902e+07	4.08e-95 ***
p	2.632109e-04	1.392386e-05	2.359206e-04	2.905011e-04	2.75e-60 ***
q	1.029977e-03	4.807957e-05	9.357425e-04	1.124211e-03	2.25e-72 ***
a1	6.183605e-01	5.196127e-02	5.165183e-01	7.202027e-01	7.21e-29 ***
b1	-3.659127e-02	1.902144e-04	-3.696408e-02	-3.621845e-02	0.00e+00 ***
c1	4.033442e+01	8.691148e-01	3.863099e+01	4.203785e+01	1.16e-181 ***
a2	3.070774e+02	4.551938e-01	3.061852e+02	3.079695e+02	0.00e+00 ***
b2	-1.609832e-02	8.328840e-04	-1.773075e-02	-1.446590e-02	2.58e-62 ***
c2	2.206358e+00	3.226597e-02	2.143118e+00	2.269598e+00	4.37e-253 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error 24243.91 on 491 degrees of freedom
Multiple R-squared: 0.999937 Residual sum of squares: 288593620294

Figure 2.5

The split-period models provide a clearer description of the two lifecycle stages. In the first 500 observations, $p = 0.000263$ and $q = 0.001030$, so $q/p \approx 3.91$. In the latter period, $p = 0.000637$ and $q = 0.004588$, giving $q/p \approx 7.20$.



Residuals:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	-182963	-19152	3316	7860	39294	131789

Coefficients:

	Estimate	Std. Error	Lower	Upper	p-value
m	1.506351e+07	9.155964e+04	1.488406e+07	1.524297e+07	0.00e+00 ***
p	6.372815e-04	4.291998e-06	6.288693e-04	6.456937e-04	0.00e+00 ***
q	4.588263e-03	4.656583e-05	4.496996e-03	4.679531e-03	0.00e+00 ***
a1	3.460302e+02	1.764826e-01	3.456843e+02	3.463761e+02	0.00e+00 ***
b1	-4.201114e-02	1.070781e-03	-4.410983e-02	-3.991245e-02	3.80e-156 ***
c1	8.859015e+00	1.384814e-01	8.587597e+00	9.130434e+00	8.72e-246 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error 66223.11 on 510 degrees of freedom
Multiple R-squared: 0.999768 Residual sum of squares: 2.236605e+12

Figure 2.6

Both models have in-sample R^2 values above 0.9997, but the later model has a larger residual standard error because it includes the exceptionally large DLC release peak.

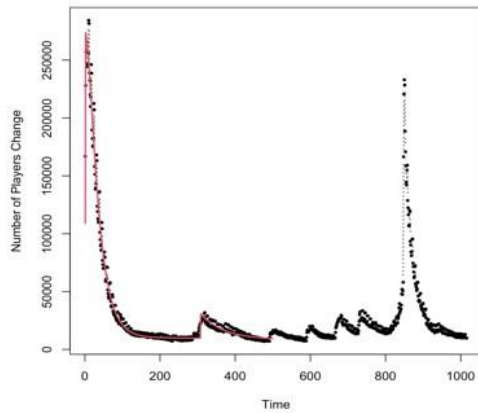


Figure 2.7

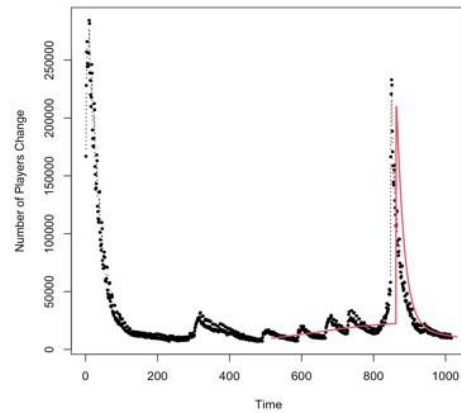


Figure 2.8

The prediction model accurately fits the first half of the data, capturing two significant peaks. However, the second half of the data shows several fluctuations before reaching the peak, which presents challenges for the model. Despite these difficulties, the model represents an increasing trend that closely predicts the overall tendency of player numbers. There are several fluctuations in mid stage of the dataset, due to the platform Steam's promotion and seasonal variations of price, which can improve a video game's sale amount.

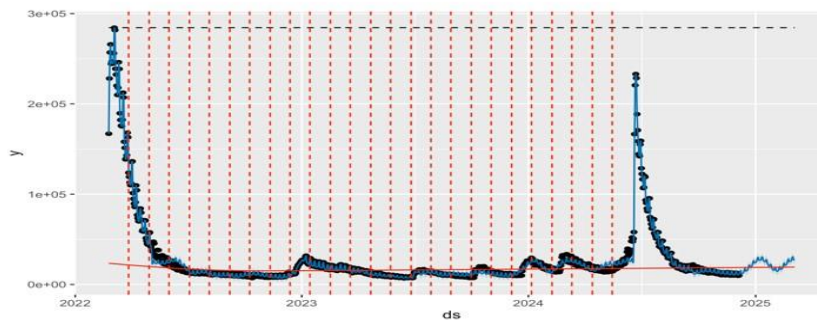


Figure 3.9

The selected Prophet model forecasts a post-DLC baseline of approximately 26,800 to 30,800 during the final six dates of the 90-day horizon, with uncertainty intervals of roughly 23,200 to 34,400. The forecast shows that activity remains well above the pre-DLC low point but far below the expansion peak. This is consistent with a strong but temporary reactivation effect followed by gradual normalization.

Sid Meier's Civilization VI

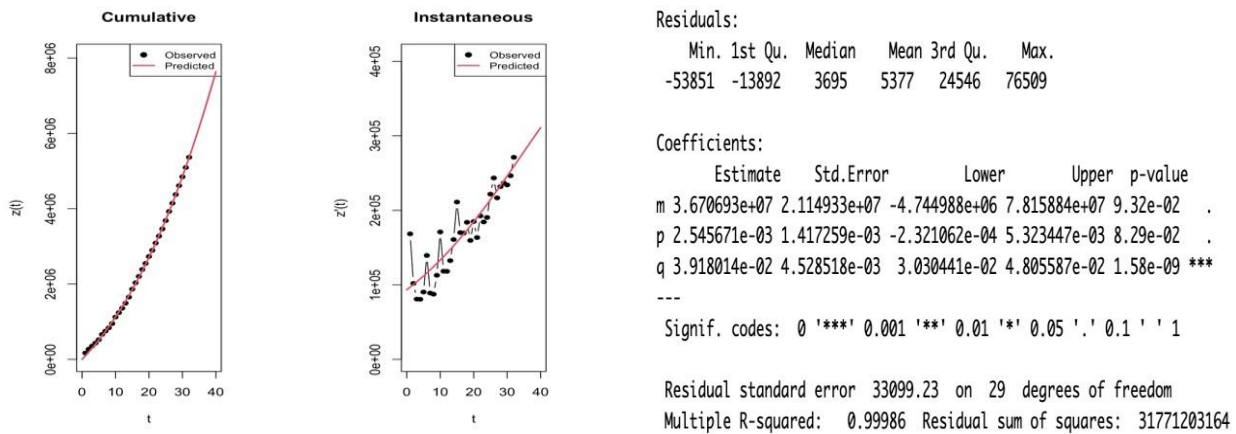


Figure 2.10

Civilization VI follows a different lifecycle from the two launch-driven games. Quarterly activity declined during the early period and then developed a sustained upward trend, rising from lows of roughly 80,000 to approximately 270,000 by the end of the sample. The Bass Model captures the cumulative curve closely, with an in-sample R^2 of 0.99986. The estimated imitation coefficient is $q = 0.03918$, around 15.4 times the innovation coefficient $p = 0.002546$. Although the Bass Model effectively predicts the general trend of the dataset, several outliers in the instantaneous plot do not align with the model's predictions.

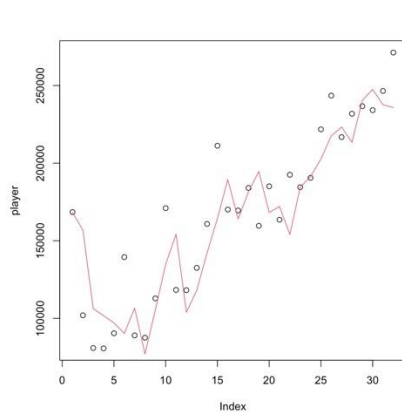


Figure 2.11

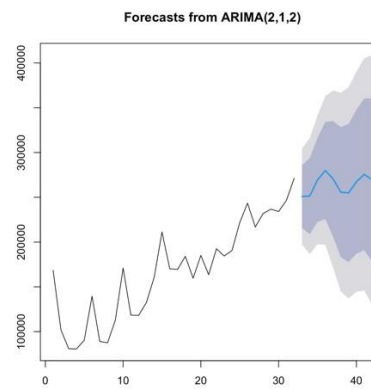


Figure 2.12

The figures represent the ARIMA model fit and forecast for player data. The model closely follows most data points, attempting to capture significant events within the dataset. Each point is the player counts quarterly, there are three points are outliers the red line does not approach exactly, this substantial changes because of the game

developer Firaxis Games released three main DLC on the specific dates during these periods.

The ARIMA(2,1,2) forecast continues the upward trajectory over the next four quarters, with central forecasts broadly between 250,000 and 280,000. The prediction intervals widen rapidly, reaching an upper range above 400,000 and a lower range near 130,000 by the end of the horizon. The model therefore supports continued high engagement but also shows substantial uncertainty about the magnitude of future quarterly activity.

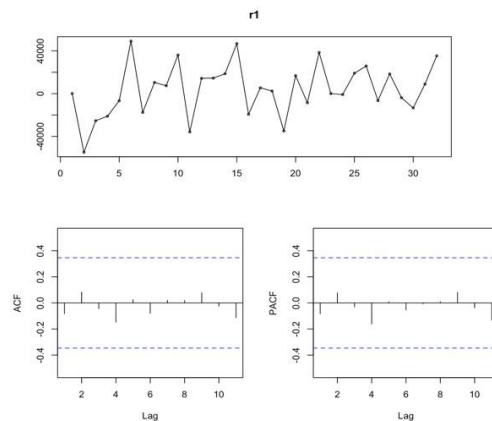


Figure 2.13

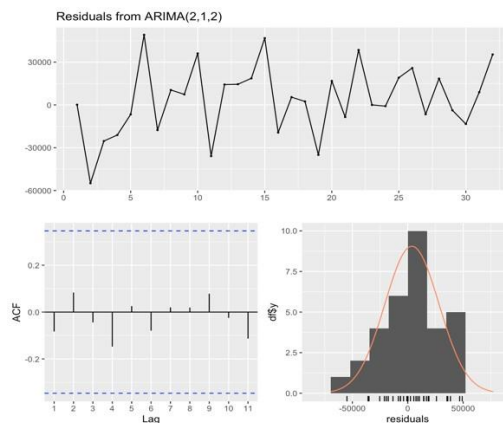


Figure 2.14

Residual diagnostics do not reveal significant remaining autocorrelation. The ACF and PACF values remain within the confidence limits, and the Ljung–Box test gives $p = 0.6333$. Therefore, there is insufficient evidence to reject the null hypothesis of no residual autocorrelation. The residual histogram is approximately bell-shaped, although some asymmetry remains.

```
Series: cumsum(player)
Regression with ARIMA(2,1,2)(1,0,1)[4] errors
```

Coefficients:

	ar1	ar2	ma1	ma2	sar1	sma1	xreg
	-0.1026	-0.9068	0.3178	1.0000	-0.0374	0.0406	0.9961
s.e.	0.1152	0.1000	0.1216	0.1545	NaN	NaN	0.0210

```
sigma^2 = 409273344: log likelihood = -349.05
AIC=714.1 AICc=720.65 BIC=725.58
```

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	-991.9274	17520.13	13947.28	-0.4102198	1.210877	0.08320614	-0.04071275

The Bass-plus-SARMAX refinement follows quarter-to-quarter variation more closely than the smooth Bass curve. Its in-sample error measures are RMSE = 17,520, MAE = 13,947, and MAPE = 1.21%. The fitted regression coefficient on the Bass component is 0.9961, showing that the cumulative Bass trajectory remains the dominant component, while the ARIMA errors adjust short-term deviations.

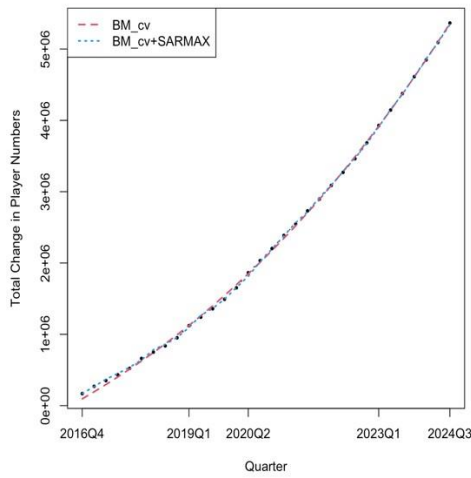


Figure 2.15

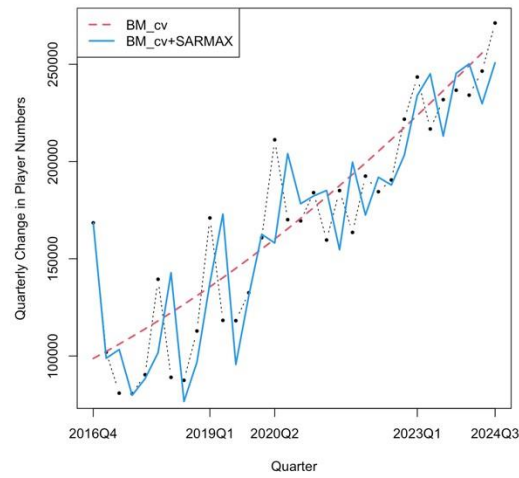


Figure 2.16

The improvement is most visible in the instantaneous plot, where the combined model follows several local peaks and declines that the Bass curve smooths over. In the cumulative plot, the two models are almost indistinguishable, so the refinement adds value mainly for short-term quarterly variation rather than the overall long-run direction.

Discussion

The three games demonstrate that successful player retention can be achieved through different lifecycle strategies. Baldur's Gate 3 shows the potential of Early Access to build community participation and launch momentum. The very high first-half q/p ratio is consistent with strong internal diffusion during the period when reviews, streaming, player discussion, and community feedback were most visible. This strategy is attractive for studios with a strong product and the capacity to respond quickly, but it can create negative effects when an unfinished product generates poor public impressions.

Elden Ring shows a different strategy. Shadow of the Erdtree generated a transformed peak approximately 81% as large as the original launch peak, and the source activity measure increased by about 301% over three days immediately before and after release. This indicates that a major expansion can reactivate dormant players at near-launch scale. The strategy is especially suitable for established franchises in which players are emotionally invested in the world, mechanics, and brand identity.

Civilization VI represents a long-tail ecosystem. Its quarterly activity recovered from early lows and increased toward the end of the sample, while the Bass and SARMAX results show a strong long-run trajectory with short-term quarterly deviations. Continuous expansions, modding, community knowledge, mature-product pricing, and replay ability are all plausible contributors. The recommended strategy is therefore not a single large event but a sustained portfolio of content, community support, and accessible product packaging.

Across all three games, the imitation coefficient exceeds the innovation coefficient. This does not prove that marketing is unimportant, but it consistently shows that player communities and existing users are central to the observed diffusion patterns. Developers should therefore treat community management, reviews, creator ecosystems, and post-launch communication as strategic commercial assets rather than secondary promotional activities.

Conclusion

This report demonstrates three games with different categories and background, highlighting how developer strategies such as early access for "Baldur's Gate 3" (BG-3), DLC releases for "Elden Ring" (EDR), and robust community engagement for "Sid Meier's Civilization VI" (CV-6) have significantly influenced their popularity and player dynamics. Although this report did not involve the topic about direct or indirect competition from other games, it is evident appeared factor, the interactions between players play a crucial role in shaping game adoption. The significance of the imitation coefficient in our models underscores the impact of existing players in attracting new ones, emphasizing the importance of community and shared players' experiences in driving game popularity.

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Guidolin, M. (2024). *Innovation Diffusion Models Theory and Practice*: Wiley.

Codes

Baldur's Gate 3

```
library(readxl) library(prophet)
library(DIMORA)
library(forecast)
```

```
BG3<- read_excel("Baldur's Gate 3 _Players.xls")
Players = BG3$Players * 0.1343096 player <-
ts(Players, frequency = 1, start = 1)
```

```
BG3_price<-read_excel("Baldur's Gate 3 _Price.xlsx") price_change
<- BG3_price$`Final price`
```

```
par(mar=c(5, 4, 4, 6) + 0.1) # Adjust the last value for the right margin as needed
```

```
# Plot for player changes
plot(player, type="o", col="#ADD8E6", xlab="Time", ylab="Number of Players Change",
main="Player Numbers and Price Fluctuations Over Time", pch=16, cex=1.2, lty=1,
ylim=c(0, max(player)*1.1))
```

```
par(new=TRUE)
plot(price_change, type="o", col="#F08080", axes=FALSE, xlab="", ylab="", pch=17, cex=1.2,
lty=2, ylim=c(0, max(price_change)*1.1)) axis(side=4, col="#F08080", col.axis="#F08080")
mtext("Price Fluctuation", side=4, line=3, col="#F08080")
```

```
legend("topright", inset=c(-0.004,0), legend=c("Number of Players Change", "Price
Fluctuation (€)"),
      col=c("#ADD8E6", "#F08080"), lty=1:2, pch=16:17, cex=0.8, pt.cex=1.5, bty="n",
xpd=TRUE)
```

```
####prediction
```

```
bm_bg <- GBM(player,shock = "exp",nshock = 2, prelmestimates = c(1.088884e+07, 0.003 ,
0.1, 11, -0.9, 1, 152, -0.5, 1))
summary(bm_bg)
```

```
####prediction
```

```
pred_bmbg<- predict(bm_bg, newx=c(1:492))
pred.instbg<- make.instantaneous(pred_bmbg)
```

```

###plot of fitted model
plot(cumsum(player), type= "b",xlab="Time", ylab="Total Change in Player Numbers",
pch=16, lty=1, cex=0.8, xlim=c(1,492), ylim=c(0,1.2e+7)) lines(pred_bmbg, lwd=2,
col=2)

plot(player, type= "b",xlab="Time", ylab="Number of Players Change", pch=16, lty=3,
cex=0.6, xlim=c(1,492))
lines(pred.instbg, lwd=2, col=2)

###estimate the model with 50% of the data
bm_bg50<-GBM(player[1:246],shock = "exp",nshock = 2, prelimestimates =
c(1.088884e+07, 2.729707e-04 , 1.270825e-02, 11, -0.9, 1, 152, -3.892836e-02, -
8.554876e-01))
summary(bm_bg50)

pred_bmbg50<- predict(bm_bg50, newx=c(1:246))
pred.instbg50<- make.instantaneous(pred_bmbg50)

plot(player, type= "b",xlab="Time", ylab="Number of Players Change", pch=16, lty=3,
cex=0.6)
lines(pred.instbg50, lwd=2, col=2)

# Prophet
Average_player <- data.frame(player, BG3$DateTime, cap=118000)
colnames(Average_player)=c("y","ds","cap")
plot(Average_player$y, x=Average_player$ds, ylab = "Number of Players Change", xlab =
"Day")

shock_dates <- data.frame(
  holiday = c('shock_event_1', 'shock_event_2'),
  ds = as.Date(c('2023-08-13', '2024-01-07')), # Example shock dates
  lower_window = c(-7,-20),
  upper_window = c(35,35) # Duration of the effect
)

m = prophet(Average_player, growth = "logistic", holidays = shock_dates) summary(m)

###create a future 'window' for prediction
future <- make_future_dataframe(m, periods = 90, freq="day", include_history = T)
tail(future)
future$cap=118000

```

```

forecast <- predict(m, future)
tail(forecast[c("ds", "yhat", "yhat_lower", "yhat_upper")])

plot(m, forecast)

#Dynamic plot
dyplot.prophet(m, forecast)

#plot with change points

plot(m, forecast)+add_changepoints_to_plot(m, threshold=0)

#dates corresponding to change points m$changepoints

##Prophet with no change points and multiplicative seasonality
m2=prophet(Average_player, growth="logistic", n.changepoints=0, holidays = shock_dates,
seasonality.mode='multiplicative')

summary(m2) m2$seasonalities

future2 <- make_future_dataframe(m2, periods = 90, freq="day", include_history = T)
tail(future2)
future2$cap=118000

forecast2 <- predict(m2, future2)
tail(forecast2[c("ds", "yhat", "yhat_lower", "yhat_upper")])

#prediction plot

plot(m2, forecast2)
dyplot.prophet(m2, forecast2)

```

```
library(readxl) library(prophet)
library(DIMORA)
library(forecast)
```

```
Edr<- read_excel("Eleden Ring Players.xls") Players
= Edr$Players * 0.2982192
player <- ts(Players, frequency = 1, start = 1)
```

```
Edr_price <- read_excel("EDR Price.xlsx")
price_change <- Edr_price$`Final price`
```

```
par(mar=c(5, 4, 4, 6) + 0.1) # Adjust the last value for the right margin as needed
```

```
# Plot for player changes
plot(player, type="o", col="#ADD8E6", xlab="Time", ylab="Number of Players Change",
main="Player Numbers and Price Fluctuations Over Time", pch=16, cex=1.2, lty=1,
ylim=c(0, max(player)*1.15))
```

```
par(new=TRUE)
plot(price_change, type="o", col="#F08080", axes=FALSE, xlab="", ylab="", pch=17, cex=1.2,
lty=2, ylim=c(0, max(price_change)*1.15)) axis(side=4, col="#F08080", col.axis="#F08080")
mtext("Price Fluctuation", side=4, line=3, col="#F08080")
```

```
legend("topright", inset=c(-0.004,0), legend=c("Number of Players Change", "Price
Fluctuation (€)"),
col=c("#ADD8E6", "#F08080"), lty=1:2, pch=16:17, cex=0.8, pt.cex=1.5, bty="n",
xpd=TRUE)
```

```
###prediction
bm_edr <- GBM(player,shock = "exp",nshock = 2, prelmestimates = c(2.803218e+07,
7.386279e-05, 1.797903e-03, 10, -5.023506e-02, 6.997694e+01, 850, -2.512700e-02,
1.142238e-02))
summary(bm_edr)
```

```
###prediction
pred_bmedr<- predict(bm_edr, newx = (1:1016))
pred.instedr<- make.instantaneous(pred_bmedr)
```

```
###plot of fitted model
plot(cumsum(player), type= "b",xlab="Time", ylab="Total Change in Player Numbers",
pch=16, lty=1, cex=0.8) lines(pred_bmedr, lwd=2, col=2)
```

```
plot(player, type= "b",xlab="Time", ylab="Number of Players Change", pch=16, lty=3,
cex=0.6)
lines(pred.instedr, lwd=2, col=2)
```

```
###estimate the model with 50% of the data
```

```
bm_edr50<-GBM(player[1:500],shock = "exp",nshock = 2, prelmestimates =
c(2.8600004e+07, 2.550158e-05, 8.935422e-04, 10, -3.027675e-02, 3.701211e+00, 319, -
0.001, 0.1))
summary(bm_edr50)
```

```
pred_bmedr50<- predict(bm_edr50, newx=c(1:500)) pred.instedr50<-
make.instantaneous(pred_bmedr50)
```

```
plot(player, type= "b",xlab="Time", ylab="Number of Players Change", pch=16, lty=3,
cex=0.6)
lines(pred.instedr50, lwd=2, col=2)
```

```
###estimate the model with 50% to 100% of the data
```

```
bm_edr50_100<-GBM(player[501:1016],shock = "exp",nshock = 1, prelmestimates =
c(2.8600004e+07, 6.389492e-04, 4.552686e-03, 350, -4.142344e-02, 8.827355e+00))
summary(bm_edr50_100)
```

```
pred_bmedr50_100<- predict(bm_edr50_100, newx=c(1:516)) pred.instedr50_100<-
make.instantaneous(pred_bmedr50_100)
```

```
x_values <- 1:length(pred.instedr50_100) adjusted_x
<- x_values + 516
plot(player, type= "b", xlab="Time", ylab="Number of Players Change", pch=16, lty=3,
cex=0.6)
lines(adjusted_x, pred.instedr50_100, lwd=2, col=2)
```

```
# Prophet
```

```
Average_player <- data.frame(player, Edr$DateTime, cap=284330)
colnames(Average_player)=c("y","ds","cap")
plot(y = Average_player$y, x = Average_player$ds, type = "o",
xlab = "Day", ylab = "Number of Players Change", lwd = 2, pch = 16, cex = 1.2)
```

```
shock_dates <- data.frame(
holiday = c('shock_event_1', 'shock_event_2', 'shock_event_3'),
ds = as.Date(c('2022-03-05', '2023-01-08', '2024-06-22')), # Example shock dates
lower_window = c(-10,-18,-3),
upper_window = c(60,180,90) # Duration of the effect
```

)

```
m = prophet(Average_player, growth = "logistic", holidays = shock_dates) summary(m)
```

```
future <- make_future_dataframe(m, periods = 90, freq="day", include_history = T)
tail(future)
future$cap=284330
```

```
forecast <- predict(m, future)
tail(forecast[c("ds", "yhat", "yhat_lower", "yhat_upper")])
```

```
plot(m, forecast)
```

```
#Dynamic plot
dyplot.prophet(m, forecast)
```

```
#plot with change points
```

```
plot(m, forecast) +add_changepoints_to_plot(m, threshold=0)
```

```
#dates corresponding to change points m$changepoints
```

```
##Prophet with no change points and multiplicative seasonality
m2=prophet(Average_player, growth="logistic", n.changepoints=0, holidays = shock_dates,
seasonality.mode='multiplicative')
```

```
summary(m2)
m2$seasonalities
```

```
future2 <- make_future_dataframe(m2, periods = 90, freq="day", include_history = T)
tail(future2)
future2$cap=284330
```

```
forecast2 <- predict(m2, future2)
tail(forecast2[c("ds", "yhat", "yhat_lower", "yhat_upper")])
```

```
#prediction plot
```

```
plot(m2, forecast2) dyplot.prophet(m2,
forecast2) Sid Meier's Civilization
library(readxl) library(DIMORA)
library(forecast)
```

```

library(tseries)

cv<- read_excel("Sid Meier's Civilization VI.xlsx")
Players = cv$Players * 0.04366329 player <-
ts(Players, frequency = 1, start = 1) player <-
player[-length(player)]

price <- cv$`Final Price`
price <- price[-length(price)]

quarter <-cv$Quarter
quarter <- quarter[-length(quarter)]

par(mar=c(5, 4, 4, 6) + 0.1) # Adjust the last value for the right margin as needed

# Plot for player changes
plot(player, type="o", col="#ADD8E6", xlab="Time", ylab="Number of Players Change",
main="Player Numbers and Price Fluctuations Over Time", pch=16, cex=1.2, lty=1,
ylim=c(0, max(player)*1.2))

par(new=TRUE)
plot(price, type="o", col="#F08080", axes=FALSE, xlab="", ylab="", pch=17, cex=1.2, lty=2,
ylim=c(0, max(price)*1.2))
axis(side=4, col="#F08080", col.axis="#F08080") mtext("Price
Fluctuation", side=4, line=3, col="#F08080")

legend("topright", inset=c(-0.004,0), legend=c("Number of Players Change", "Price
Fluctuation (€)"),
      col=c("#ADD8E6", "#F08080"), lty=1:2, pch=16:17, cex=0.8, pt.cex=1.5, bty="n",
xpd=TRUE)

#Bass Model
bm_cv <- BM(player, display = TRUE)

summary(bm_cv)

pred_bmcv<- predict(bm_cv, newx=c(1:32))
pred.instcv<- make.instantaneous(pred_bmcv)

plot(cumsum(player), type= "b",xlab="Time", ylab="Total Change in Player Numbers",
pch=16, lty=3, cex=0.6)
lines(pred_bmcv, lwd=2, col=2)

```

```
plot(player, type= "b",xlab="Time", ylab="Quarterly Change in Player Numbers", pch=16,
lty=3, cex=0.6)
lines(pred.instcv, lwd=2, col=2)
```

```
####Arima model
```

```
a1<- Arima(player, order=c(2,1,2), seasonal=c(1,0,1)[4])
fit1<- fitted(a1) summary(fit1)#!
```

```
plot(player)
lines(fit1, col=2)
```

```
f1<- forecast(a1) plot(f1)
```

```
r1<- residuals(a1)
tsdisplay(r1)
```

```
checkresiduals(f1)
```

```
##Exploratory plots
```

```
plot(player, type= "b",xlab="Quarter", ylab="Quarterly Change in Player Numbers", pch=16,
lty=3, cex=0.6, xaxt="n")
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
```

```
plot(cumsum(player), type= "b",xlab="Quarter", ylab="Total Change in Player Numbers",
pch=16, lty=3, cex=0.6, xaxt="n")
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
```

```
##BM
```

```
bm_cv<- BM(player, display = TRUE)
```

```
##Prediction with pred_bm_cv2 pred_bm_cv<-
predict(bm_cv, newx=c(1:32)) pred_bm_cv_i<-
make.instantaneous(pred_bm_cv)[-1]
```

```
##Plots observed vs predicted
```

```
plot(player, type= "b",xlab="Quarter", ylab="Quarterly Change in Player Numbers", pch=16,
lty=3, cex=0.6, xaxt="n", col=1)
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
lines(pred_bm_cv_i, lwd=2, col=2)
```

```

plot(cumsum(player), type= "b",xlab="Quarter", ylab="Total Change in Player Numbers",
pch=16, lty=3, cex=0.6, xaxt="n",col=1)
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
lines(pred_bm_cv, lwd=2, col=2)

##SARMAX refinement
fit.player<- fitted(bm_cv)
s2 <- Arima(cumsum(player), order = c(2,1,2), seasonal=list(order=c(1,0,1), period=4), xreg
= fit.player)
summary(s2)

pres2 <- make.instantaneous(fitted(s2))

##Plots observed vs predicted with SARMAX refinement
plot(player, type= "b",xlab="Quarter", ylab="Quarterly Change in Player Numbers", pch=16,
lty=3, cex=0.6, xaxt="n", col=1)
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
lines(pred_bm_cv_i, lty=2, lwd=2, col=2) lines(pres2, lty=1,lwd=2, col=4)
legend("topleft", legend=c("BM_cv","BM_cv+SARMAX"), lty=c(2,1), lwd = c(2,2), col = c(2,4))
#
plot(cumsum(player), type= "b",xlab="Quarter", ylab="Total Change in Player Numbers",
pch=16, lty=3, cex=0.6, xaxt="n",col=1)
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
lines(pred_bm_cv, lty=2, lwd=2, col=2) lines(cumsum(pres2), lty=3,
lwd=2, col=4)
legend("topleft", legend=c("BM_cv","BM_cv+SARMAX"), lty=c(2,3), lwd = c(2,2), col=c(2,4))

```